

Element B: Documentation and Analysis of Prior Design Attempts

Problem; Stakeholders; Requirements

The stakeholders, school librarian Ms. Tatro, students, and faculty, require sustainable and recyclable bookends for the library. They need to be able to sustain the weight of the books, not move and support the Green School initiative.

Understanding Bookends

Marcusa, A. (2025). *Best Bookends*. BestReviews.

<https://bestreviews.com/office/supplies/best-bookends/>

Vieve, P. (2024, March 10). *The Art of Bookends: A Comprehensive Guide*. Rubblemagazine.

<https://rubblemagazine.com/bookends/>

Steel Bookends

Stakeholder:

The current stakeholders are using steel bookends in the library right now.

It's been a problem because it's been rusting, it's old, it's too heavy, it's not sturdy enough, and it's not aesthetically pleasing.

Sources:

EDUCBA. (2023, January 10). *What are the Advantages and*

Disadvantages of Steel? EDUCBA.

<https://www.educba.com/advantages-and-disadvantages-of-steel/>

Kat de Naoum. (2023, April 21). *Steel: Definition, Composition, Types,*

Properties, and Applications. Www.xometry.com; Xometry.

<https://www.xometry.com/resources/materials/steel/>

Sabhadiya, S. (2024, June 11). *What Is Steel?- Properties, Uses | Types Of Steels*.

<https://www.theengineeringchoice.com/what-is-steel-and-types-of-steel/>



Structural Community. (2020, April 2). *Advantages & Disadvantages of Steel Construction*.

Structural Community.

<https://structuralcommunity/articles/advantages-of-steel-construction/>

View. (2025, January 28). *Pros and Cons of Steel Structures*. Rigid Academy-Building Is an Art.

<https://rigidacademy.com/2025/01/28/pros-and-cons-of-steel-structures/>

Description:

Steel as a building material for bookends are strong and renewable. Steel can be altered into multiple various forms, allowing for multiple different designs that could be made with bookends. Steel also provides strength under even tension and heavy loads, allowing for multiple books to be put onto a bookend made out of this material.



Strengths:

- Recyclable & Renewable (Eco-Friendly)
- Lots of Durability & Strength
- Can be designed in many different ways
- Long-Term Cost Efficiency

Weaknesses:

- Isn't aesthetically pleasing
- Vulnerable to rusting
- Sensitive to heat and fire
- Prone to buckling (possible to bend out of shape)

Bamboo Bookends

Sources:

ltp123. (2025, February 27). *Bamboo as a Sustainable Building Material: Pros and Cons* | *Live to Plant*. Livetoplant.com.

<https://livetoplant.com/bamboo-as-a-sustainable-building-material-pros-and-cons/>

Mueller, C. (n.d.). *The Pros and Cons of Using Bamboo as a Sustainable Building Material*.

Wood Questions. Retrieved March 20, 2025, from

<https://woodquestions.com/the-pros-and-cons-of-using-bamboo-as-a-sustainable-building-material/>

The Pros and Cons of Bamboo in Green Building. (n.d.). Elemental.green. Retrieved March 18, 2025, from <https://elemental.green/the-pros-and-cons-of-bamboo-in-green-building/>

Description:

Bamboo as a building material for bookends are renewable and sustainable. They are environmentally friendly alternatives to traditional wood bookends. They often have smooth, polished finishes that make them aesthetically pleasing while still being strong enough to hold books and lightweight enough to easily move them.



Strengths:

- Sustainable
- High strength to weight ratio
- Flexible and lightweight
- Affordable
- Natural grain patterns/aesthetics

Weaknesses:

- Deteriorable
- Moisture Sensitivity

Glass Bookends

Sources:

14 Advantages of Using Glass in Construction - BuilderSpace. (n.d.). www.builderspace.com.

Retrieved March 20, 2025, from

<https://www.builderspace.com/advantages-of-using-glass-in-construction>

Barnes, R. (2023, August 3). *Pros and Cons of Using Glass as a Building Material*.

Constructionreview.

<https://constructionreviewonline.com/knowhow/installations-materials/pros-and-cons-of-using-glass-as-a-building-material/>

Team, P. (2024, October 24). *Pros And Cons Of Glass*. PROS plus CONS.

<https://prospluscons.com/pros-and-cons-of-glass/>

Description:

Glass as a building material for bookends are aesthetically pleasing and renewable. Glass provides for versatility in many forms in tempered glass, laminated glass, and heat-resistant glass. A bookend made out of glass would be able to hold a few books.

Strengths:

- Transparency and Aesthetically Appealing
- Sustainable and Recyclable
- Heat-Resistant and Rust-Resistant
- Durable and Strong (when treated)
- Versatile
- Waterproof

Weaknesses:

- Easily fragile
- Has a heavy weight
- Poor impact resistance
- Has limited flexibility
- Can easily be scratched



Application To New Solution:

Due to all the strengths and weaknesses of the three previous bookends and our stakeholder's current bookends, a bookend that's made out of wood would be the best bookend to implement. They have the least amount of weaknesses with the most beneficial features that should be found within a bookend. A bookend made out of wood can be aesthetically pleasing with the possibility of painting it to be pleasing to the eye. A wooden bookend combines the strength of the steel bookend with the sustainability and light weight of bamboo bookends and the aesthetic appeal of the glass bookend. It does this without having the weakness of rusting like the steel bookend, deterioration of the bamboo bookend, and fragility of the glass bookend.